

EXAM IN: STA510 STATISTICAL MODELING AND SIMULATION

DURATION: 4 HOURS

DATE: NOVEMBER 18th, 2024

PERMITTED AIDS: Approved simple calculator

THE EXAM CONSISTS OF 4 PROBLEMS ON 5 PAGES, 5 PAGES OF ENCLOSURES.

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Problem 1

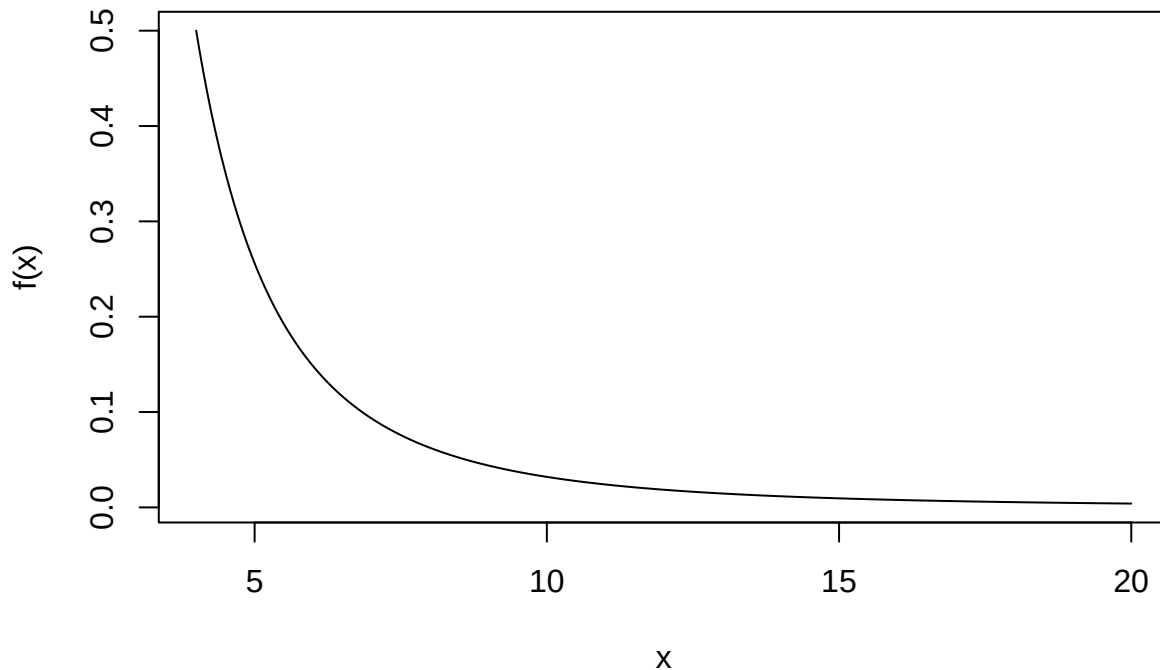
Assume that $X \sim \text{Pareto}(a, b)$ distributed with location parameter $a > 0$ and shape parameter $b > 0$. The density function $f(x)$ of X is given by

$$f(x) = \frac{ba^b}{x^{b+1}}, \quad \text{for } 0 < a < x \text{ and } 0 < b.$$

- a) Show that $f(x)$ is a proper density function. (3P)
- b) Show that the cumulative distribution function $F(x) = 1 - \left(\frac{a}{x}\right)^b$ for $0 < a < x$ and $0 < b$. (3P)
- c) Explain (with words and formulas, not code) how we can simulate $n = 10000$ random variables from this distribution using the inverse transformation method. (5P)

```
r.pareto1 <- function(n){  
  return(4/sqrt(runif(n,2,4)))  
}  
r.pareto2 <- function(n){  
  return(4/sqrt(runif(n)))  
}  
r.pareto3 <- function(n){  
  return(4/sqrt(1-runif(n)))  
}  
r.pareto4 <- function(n){  
  return(2/(1-runif(n)^(1/4)))  
}
```

- d) Given $a = 4$ and $b = 2$, which of the above functions correctly simulates random numbers with density $f(x)$? Argue why the functions produce wrong or correct results. (8P)
- e) Given $a = 4$ and $b = 2$, calculate $P(X < 6)$ and $P(10 < X < 15)$. (3P)



- f) Given $a = 4$ and $b = 2$, the density $f(x)$ is shown in the above figure. Assume we have simulated $n = 10000$ random variables and we want to create a histogram. Would you recommend to use the "Scott" rule for the automatic bin-width selection? Why? (3P)

Assume now we want to estimate the integral

$$I = \int_a^\infty \frac{\ln(x)}{x^b} dx, \quad \text{for } a > 0, b > 0.$$

- g) Explain how we can estimate I by a generalized crude Monte Carlo estimator and write down an expression for the Monte Carlo estimator based on n random draws from a $Pareto(a, b)$ distribution. (4P)
- h) Explain if and how we can estimate I by an antithetic Monte Carlo estimator. Write down an expression for the antithetic Monte Carlo estimator based on n random draws from a $Pareto(a, b)$ distribution. (5P)

Problem 2

Assume we can only generate random numbers from the standard uniform distribution $U(0, 1)$ and the standard normal distribution $N(0, 1)$.

- a) Describe how we can generate random numbers X which are Lognormal distributed with mean $\mu = 1$ and variance $\sigma^2 = \frac{1}{4}$. (3P)
- b) Describe how we can generate random numbers Y which are χ^2 -distributed with 2 degrees of freedom. (3P)
- c) Let $Z = 5 \cdot X + 3 \cdot Y$. Describe how we can find $E(Z)$, $P(Z < 20)$ and $P(40 < Z < 60)$ by $n = 10000$ Monte Carlo simulations. (4P)

d) Assume we want to estimate the integral

$$I = \int_1^3 \int_{-2}^2 \int_0^\infty \frac{x}{y^2 + 2z} e^{-\frac{1}{4}x^2} dx dy dz.$$

Write down an expression for a Monte Carlo estimator of I based on n random draws from a uniform and/or normal distribution. (4P)

Problem 3

Suppose that X is a random variable which describes the number of newborns in a given time unit.

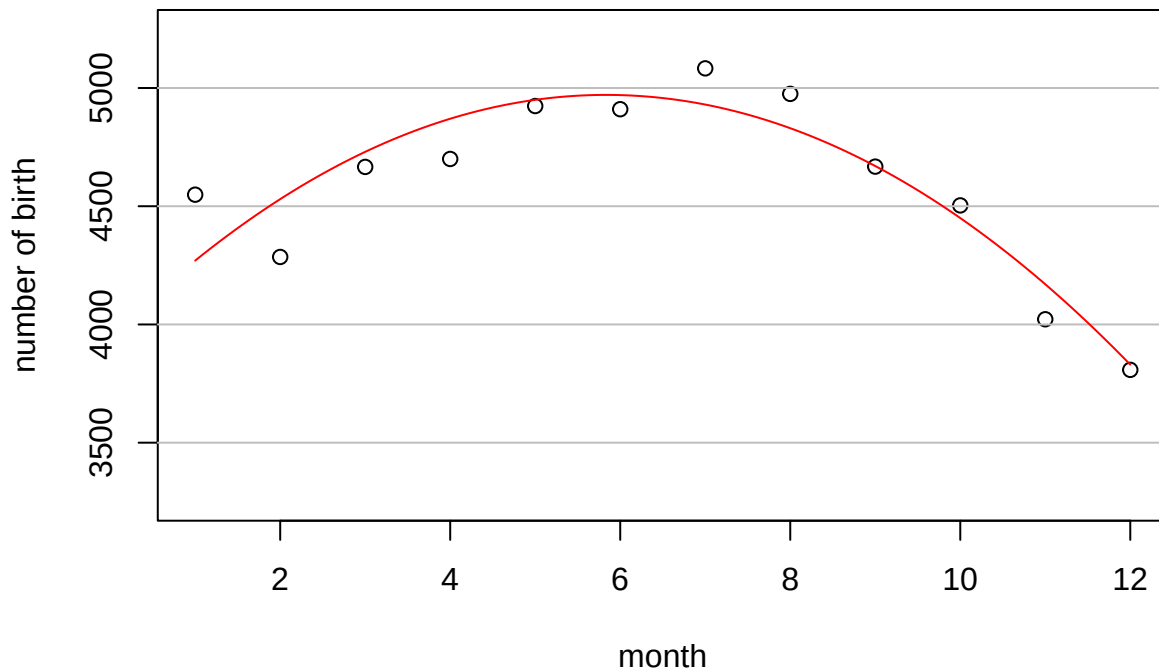
- a) Explain why its natural to model X by Poisson distribution with density function $p(x|\mu)$. (3P)

Assume the expected number of newborns at the Stavanger University hospital is 6 per day and the arrival of newborns can be modeled by a homogeneous Poisson process.

- b) What is the probability that we have less than 3 newborns at a given day? (4P)
- c) What is the probability that it takes more than 8 hours until the third baby is born on a given day? (5P)
- d) Describe one possibility how we can simulate times of birth for 30 days. (5P)

The number of newborns in Norway in a year can be modeled by a nonhomogeneous Poisson process with intensity $\lambda(t) = -30t^2 + 350t + 3950$. The below figure visualizes the average number of birth per months between 2016-2022 (small black circles) and the fitted intensity $\lambda(t)$ (red solid line).

Average of number of birth between 2016–2022



- e) What is the expected value of newborns in Norway for a year? (3P)
- f) Describe how we can simulate times of birth for 30 days by a non-homogeneous Poisson process with intensity $\lambda(t)$ (visualized by the red line in the above figure). Specify potential constants. (5P)

Problem 4

We wish to draw samples from a distribution with density $f(x)$ with

$$f(x) \propto \ln(x) * \exp\left(-\frac{(x-5)^2}{2\tau^2}\right), \quad x \geq 1, \tau > 0.$$

For this we are implementing two MCMC algorithm.

```
gx <- function(x,tau=2){
  if(x<1) res <- 0
  else res <- log(x)*exp(-0.5 * (x-5)^2 / tau^2)
  return( res )
}

# Algorithm 1
MCMC.1 <- function(gx,tau=2,sigma=1,theta1=1,n.iter=10000){
  res <- numeric(n.iter)
  res[1] <- theta1
  wt.old <- gx(res[1],tau)/dnorm(res[1],mean=5,sd=sigma)
  for(i in 2:n.iter){
    thetaStar <- rnorm(1,mean=5,sd=sigma)
    wt.star <- gx(thetaStar,tau)/dnorm(thetaStar,mean=5,sd=sigma)
    alpha <- min(1.0,wt.star/wt.old)
    if(thetaStar>=1 & runif(1)<alpha){
      res[i] <- thetaStar
      wt.old <- wt.star
    } else {
      res[i] <- res[i-1]
    }
  }
  return(res)
}
```

- a) What type of algorithm is the function MCMC.1? Explain and justify your answer! (5P)

```

# Algorithm 2
MCMC.2 <- function(gx,tau=1,sigma=1,theta1=2,n.iter=10000){
  res <- numeric(n.iter)
  res[1] <- theta1
  for(i in 2:n.iter){
    thetaStar <- rnorm(1,mean=res[i-1],sd=sigma)
    alpha <- min(1,gx(thetaStar,tau)/gx(res[i-1],tau))
    if(thetaStar>=1 & runif(1)<alpha){
      res[i] <- thetaStar
    } else {
      res[i] <- res[i-1]
    }
  }
  return(res)
}

```

- b) What type of algorithm is the function MCMC.2? Explain and justify your answer! (5P)
- c) Which of the algorithm would you prefer for the given goal? Explain you answer! (3P)
- d) Below you see a trace plot of the MCMC chain generated by the second algorithm using the call `MCMC.2(gx,tau=3,theta1=100,sigma=3,n.iter=1000)` and two values to quantify the MCMC output. Can we use the generated sample as it is, or do you suggest to improve the generated sample for the target distribution? Why and what are your suggestions? (5P)

